

LAB EXPERIMENTS

Date: 05.08.2026

Questions : 21-25

Set 21:

Q1. Histogram and Density Plot for Units Consumed

Program:

```
sector <- c("Residential", "Commercial", "Industrial",  
            "Residential", "Commercial", "Industrial")  
region <- c("North", "South", "West", "East", "North", "South")  
month <- c("Jan", "Jan", "Feb", "Feb", "Mar", "Mar")  
temperature <- c(15, 24, 20, 18, 28, 30)  
units_consumed <- c(320, 540, 880, 350, 610, 920)  
cost <- c(2100, 3600, 5900, 2300, 4100, 6200)  
renewable_usage <- c(22, 18, 12, 25, 20, 15)  
peak_hours <- c(4, 6, 8, 5, 7, 9)  
energy <- data.frame(  
  sector,  
  region,  
  month,  
  temperature,  
  units_consumed,  
  cost,  
  renewable_usage,  
  peak_hours  
)  
  
print(energy)  
  
# Histogram  
  
hist(  
  energy$units_consumed,  
  main = "Histogram of Units Consumed",  
  xlab = "Units Consumed (kWh)",  
  ylab = "Frequency",  
  col = "skyblue",  
  border = "black"  
)  
  
# Density Plot  
  
plot(  
  density(energy$units_consumed),
```

```

main = "Density Plot of Units Consumed",
xlab = "Units Consumed (kWh)",
ylab = "Density",
col = "blue",
lwd = 2
)

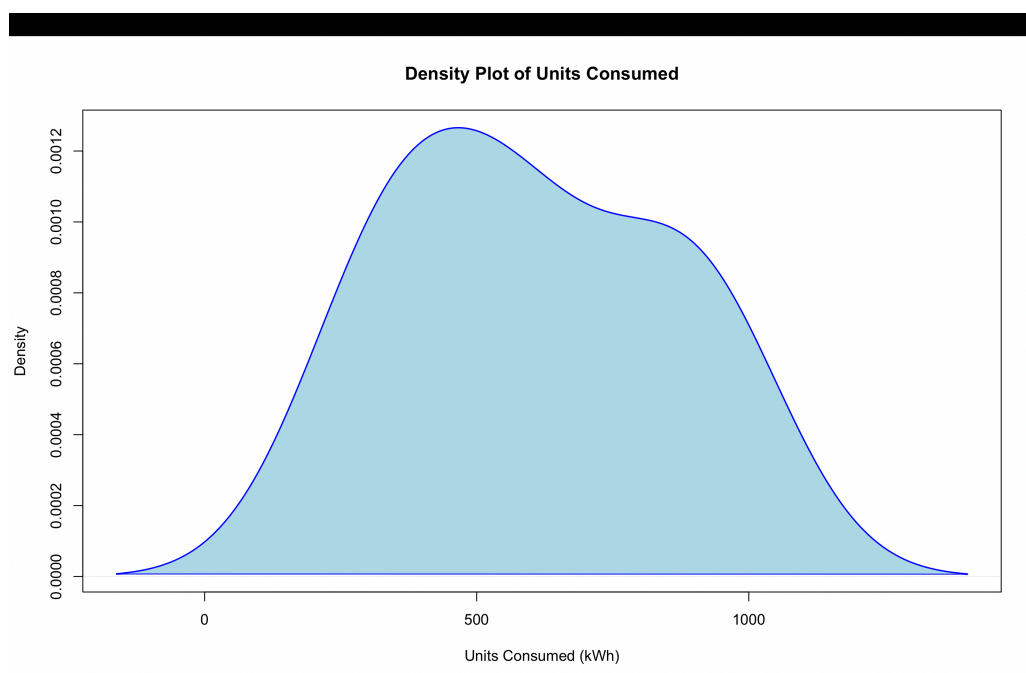
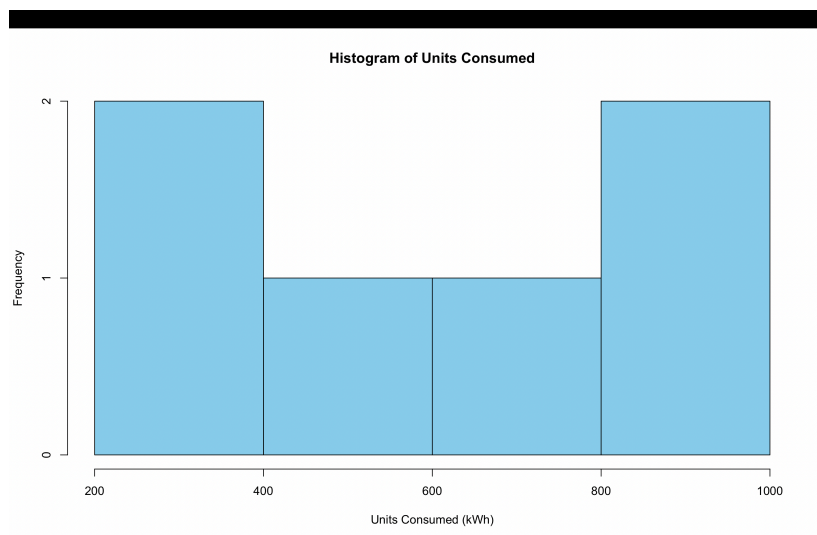
```

```

polygon(
  density(energy$units_consumed),
  col = "lightblue",
  border = "blue"
)

```

Output:



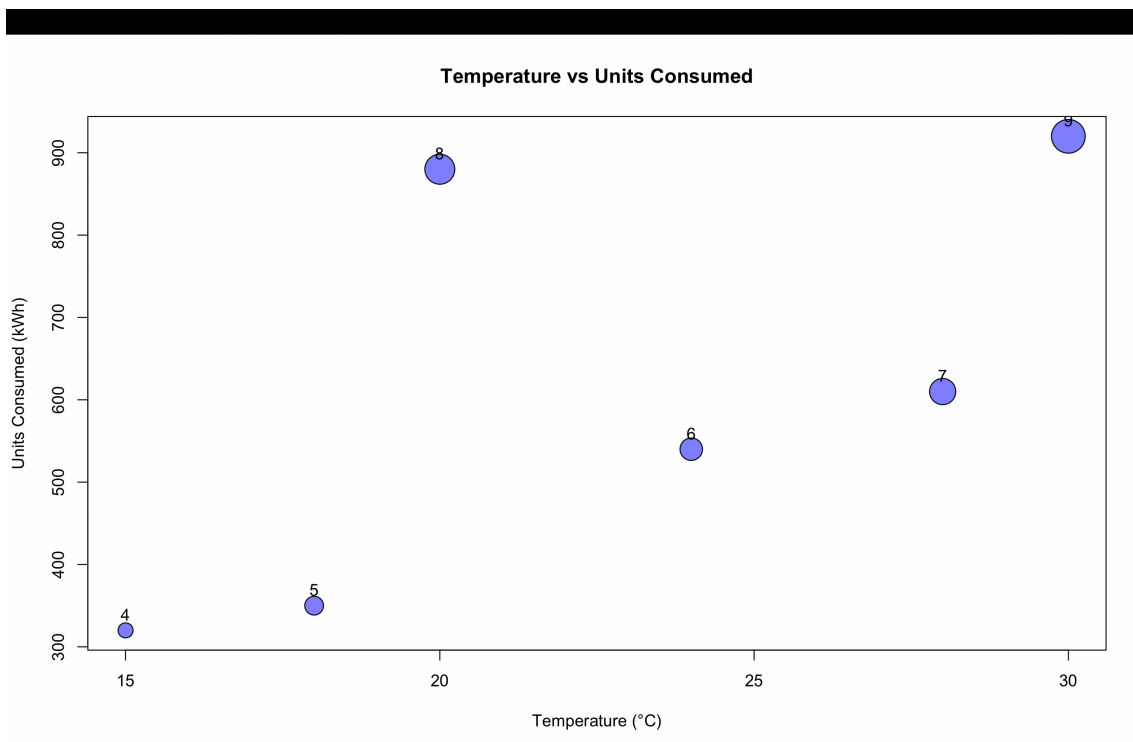
Q2. Scatter Plot Between Temperature and Units Consumed

Program:

```
plot(  
  energy$temperature,  
  energy$units_consumed,  
  main = "Temperature vs Units Consumed",  
  xlab = "Temperature (°C)",  
  ylab = "Units Consumed (kWh)",  
  pch = 21,  
  bg = rgb(0, 0, 1, 0.5),  
  cex = energy$peak_hours / 2  
)
```

```
text(  
  energy$temperature,  
  energy$units_consumed,  
  labels = energy$peak_hours,  
  pos = 3  
)
```

Output:

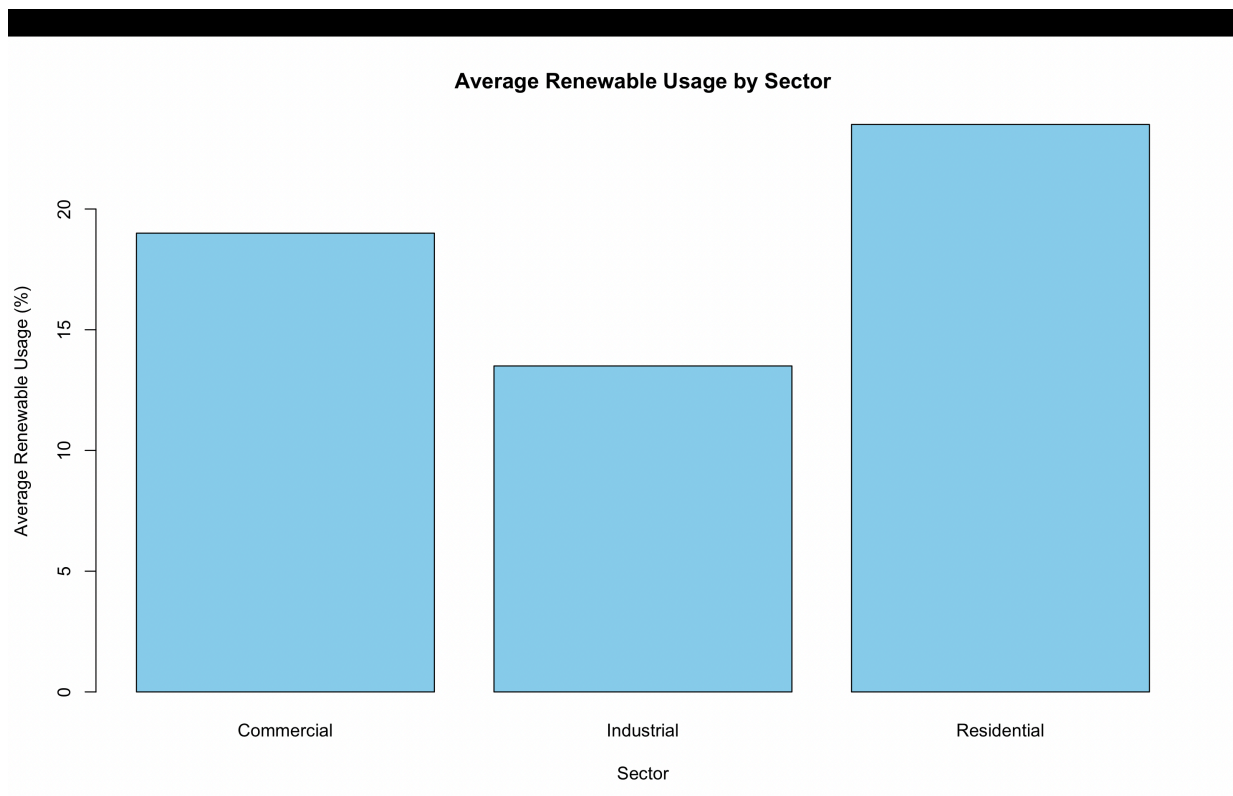


Q3. Average Renewable Usage for Each Sector

Program:

```
avg_renewable <- aggregate(  
  energy$renewable_usage,  
  by = list(energy$sector),  
  FUN = mean  
)  
  
names(avg_renewable) <- c(  
  "Sector",  
  "Average_Renewable_Usage"  
)  
  
print(avg_renewable)  
  
barplot(  
  avg_renewable$Average_Renewable_Usage,  
  names.arg = avg_renewable$Sector,  
  main = "Average Renewable Usage by Sector",  
  xlab = "Sector",  
  ylab = "Average Renewable Usage (%)",  
  col = "skyblue"  
)
```

Output:



Q4. Monthly Energy Usage Trends and Heatmap with Filters

Program:

```
library(ggplot2)

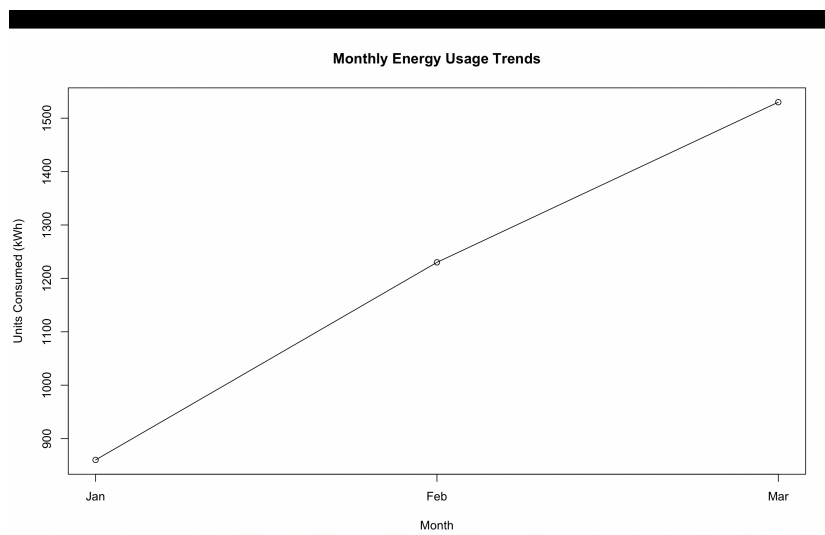
energy$month <- factor(energy$month, levels = c("Jan", "Feb", "Mar"))

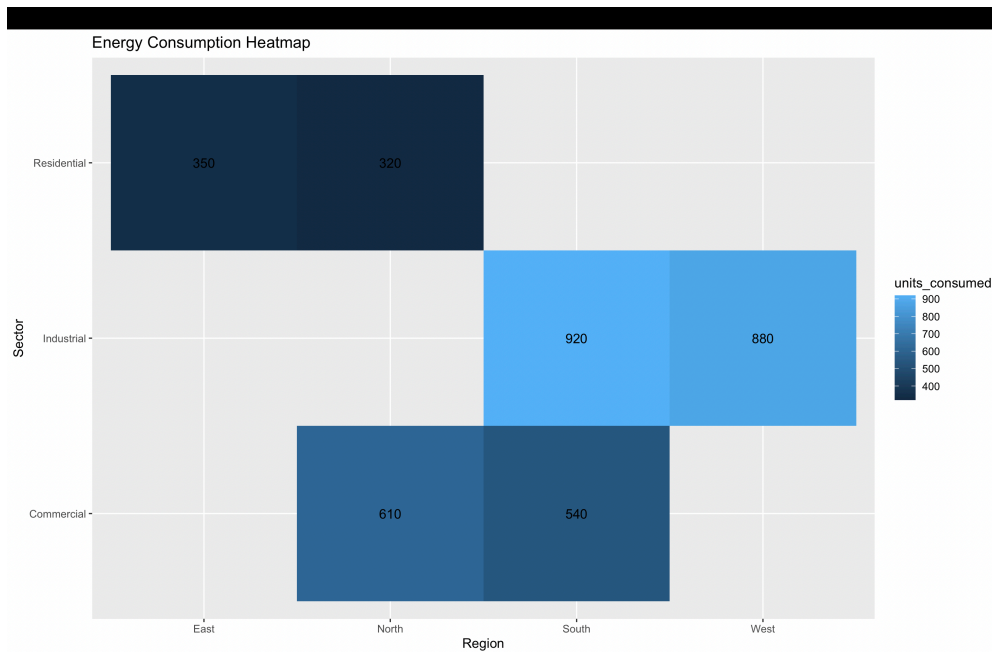
monthly <- aggregate(
  energy$units_consumed,
  list(energy$month),
  sum
)

plot(
  monthly$Group.1,
  monthly$x,
  type = "o",
  main = "Monthly Energy Usage Trends",
  xlab = "Month",
  ylab = "Units Consumed (kWh)"
)

ggplot(energy, aes(x = region, y = sector, fill = units_consumed)) +
  geom_tile() +
  geom_text(aes(label = units_consumed)) +
  labs(
    title = "Energy Consumption Heatmap",
    x = "Region",
    y = "Sector"
  )
)
```

Output:





Set 22: Time Series Analysis

Q1. Time Series Line Chart for Monthly Sales

Program:

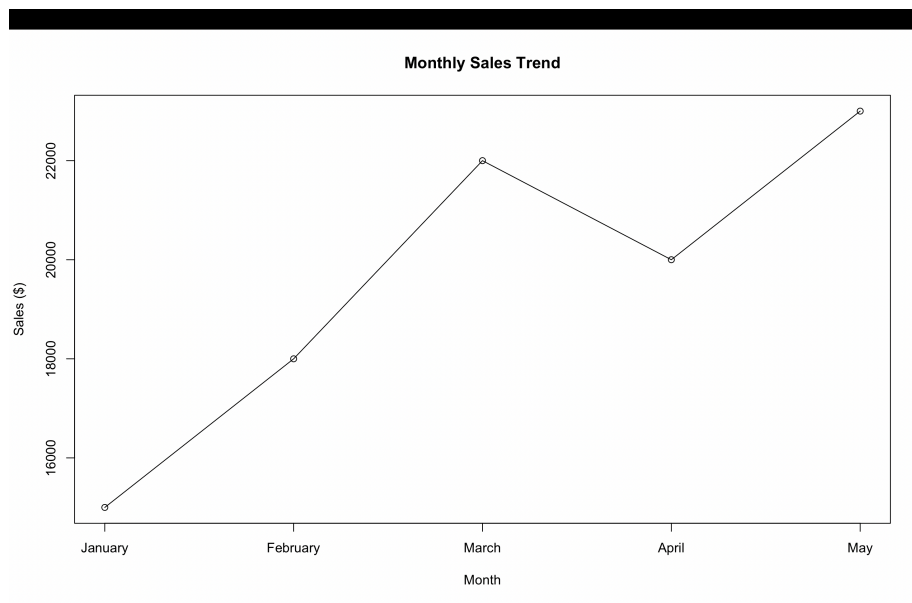
Monthly Sales Trend

```
sales <- c(15000, 18000, 22000, 20000, 23000)
```

```
plot(
  sales,
  type = "o",
  xaxt = "n",
  main = "Monthly Sales Trend",
  xlab = "Month",
  ylab = "Sales ($)"
)
```

```
axis(
  1,
  at = 1:5,
  labels = c("January", "February", "March", "April", "May")
)
```

Output:

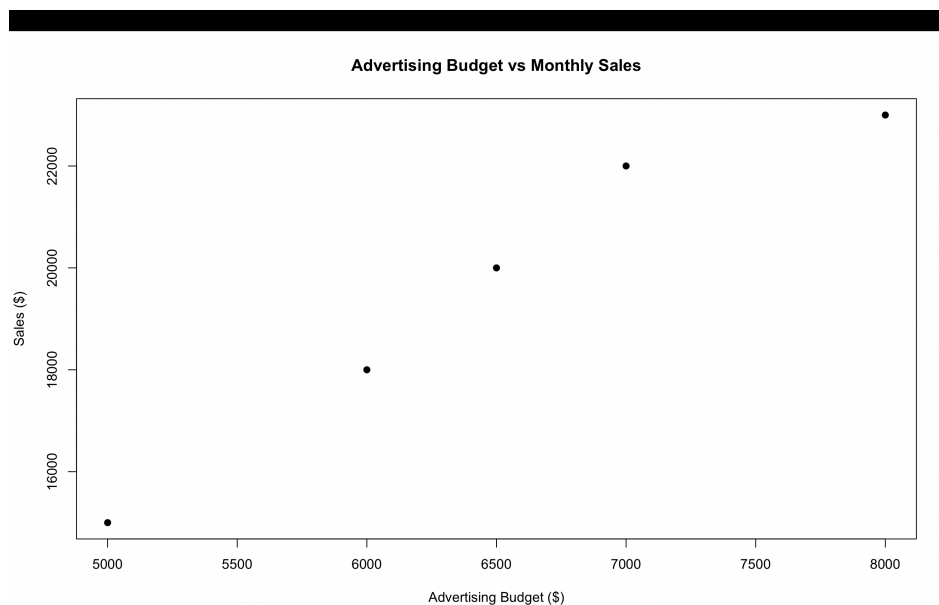


Q2. Scatter Plot Between Advertising Budget and Monthly Sales

Program:

```
advertising <- c(5000, 6000, 7000, 6500, 8000)
sales <- c(15000, 18000, 22000, 20000, 23000)
plot(
  advertising,
  sales,
  main = "Advertising Budget vs Monthly Sales",
  xlab = "Advertising Budget ($)",
  ylab = "Sales ($)",
  pch = 19
)
```

Output:

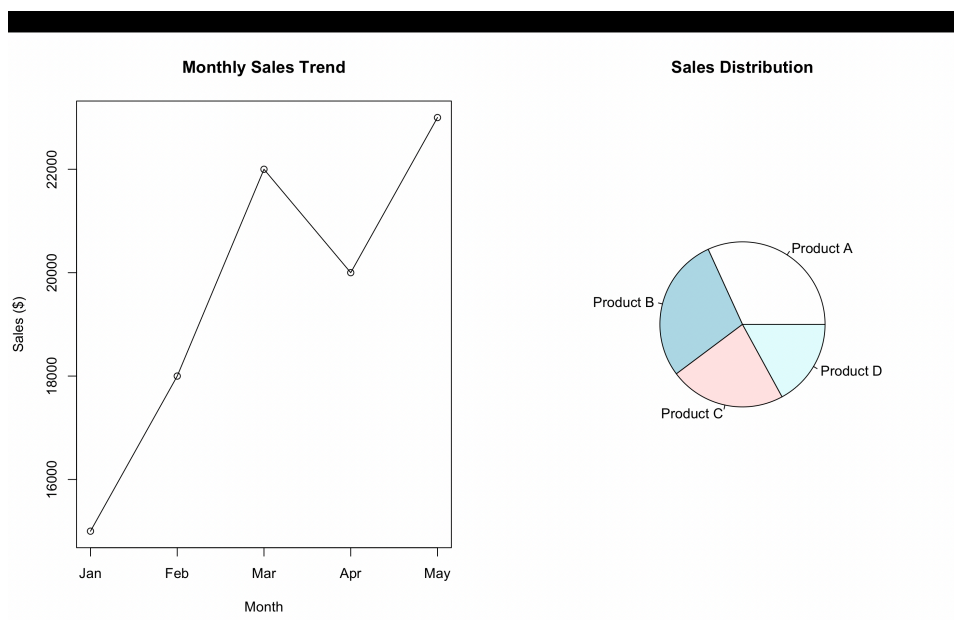


Q3. Sales Trend and Sales Distribution Dashboard

Program:

```
sales <- c(15000, 18000, 22000, 20000, 23000)
months <- c("Jan", "Feb", "Mar", "Apr", "May")
products <- c("Product A", "Product B", "Product C", "Product D")
product_sales <- c(28000, 25000, 20000, 15000)
par(mfrow = c(1, 2))
plot(
  sales,
  type = "o",
  xaxt = "n",
  main = "Monthly Sales Trend",
  xlab = "Month",
  ylab = "Sales ($)"
)
axis(1, at = 1:5, labels = months)
pie(
  product_sales,
  labels = products,
  main = "Sales Distribution"
)
par(mfrow = c(1, 1))
```

Output:



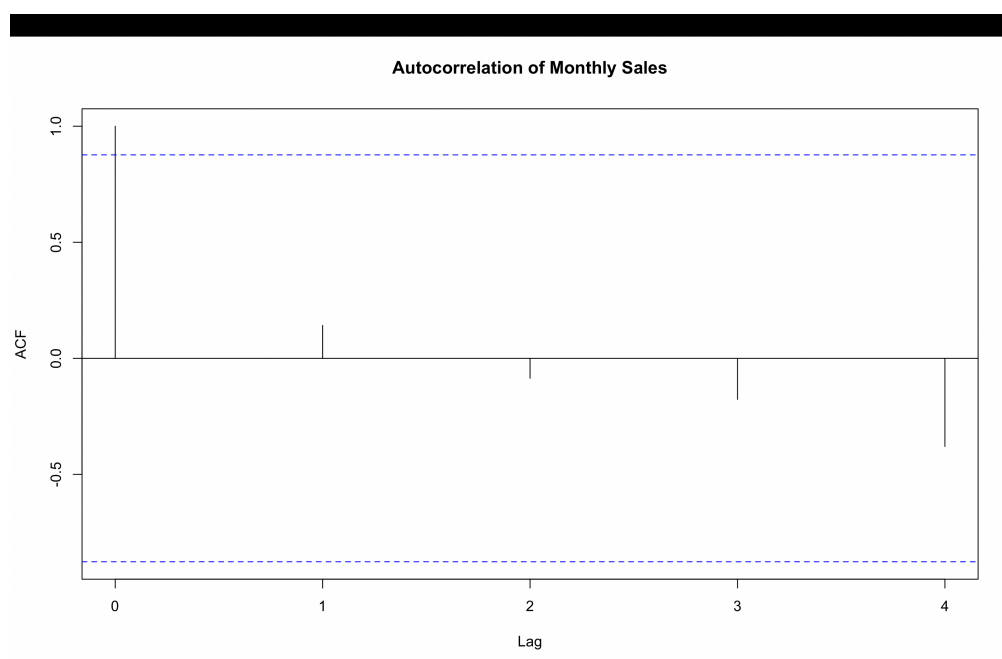
Q4. Autocorrelation Plot for Monthly Sales

Program:

```
# Autocorrelation Plot
sales <- c(15000, 18000, 22000, 20000, 23000)

acf(
  sales,
  main = "Autocorrelation of Monthly Sales"
)
```

Output:



Set 23: Employee Performance Analysis

Q1. Employee Distribution Across Departments

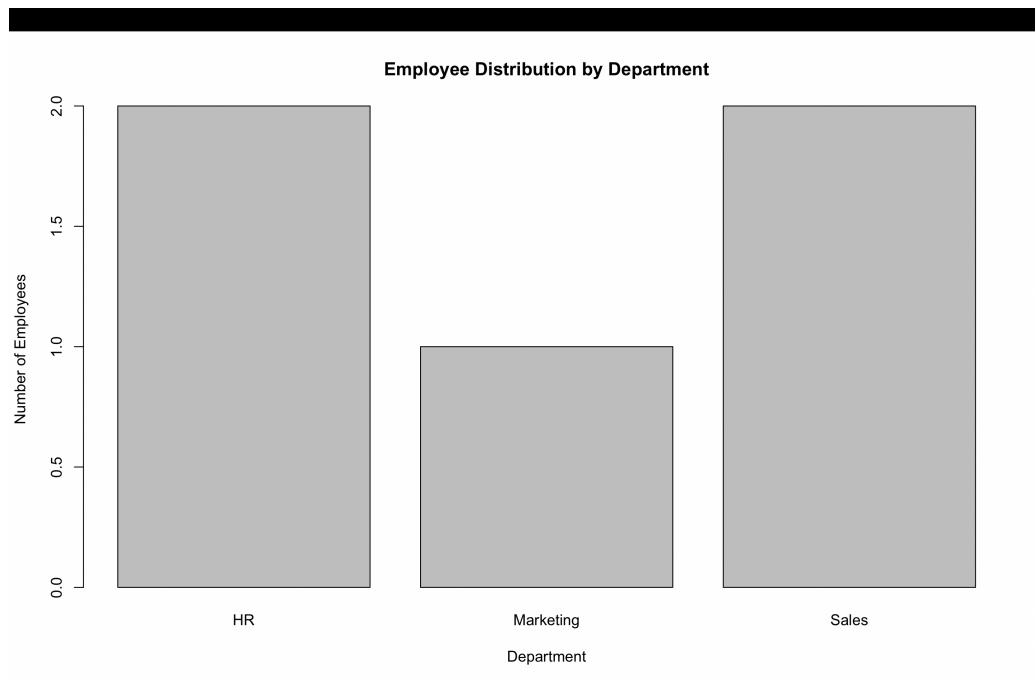
Program:

```
employee <- data.frame(
  id = c(1,2,3,4,5),
  department = c("Sales", "HR", "Marketing", "Sales", "HR"),
  years = c(5,3,7,4,2),
  performance = c(85,92,78,90,76)
)

barplot(
  table(employee$department),
  main = "Employee Distribution by Department",
  xlab = "Department",
```

```
ylab = "Number of Employees"
)
```

Output:

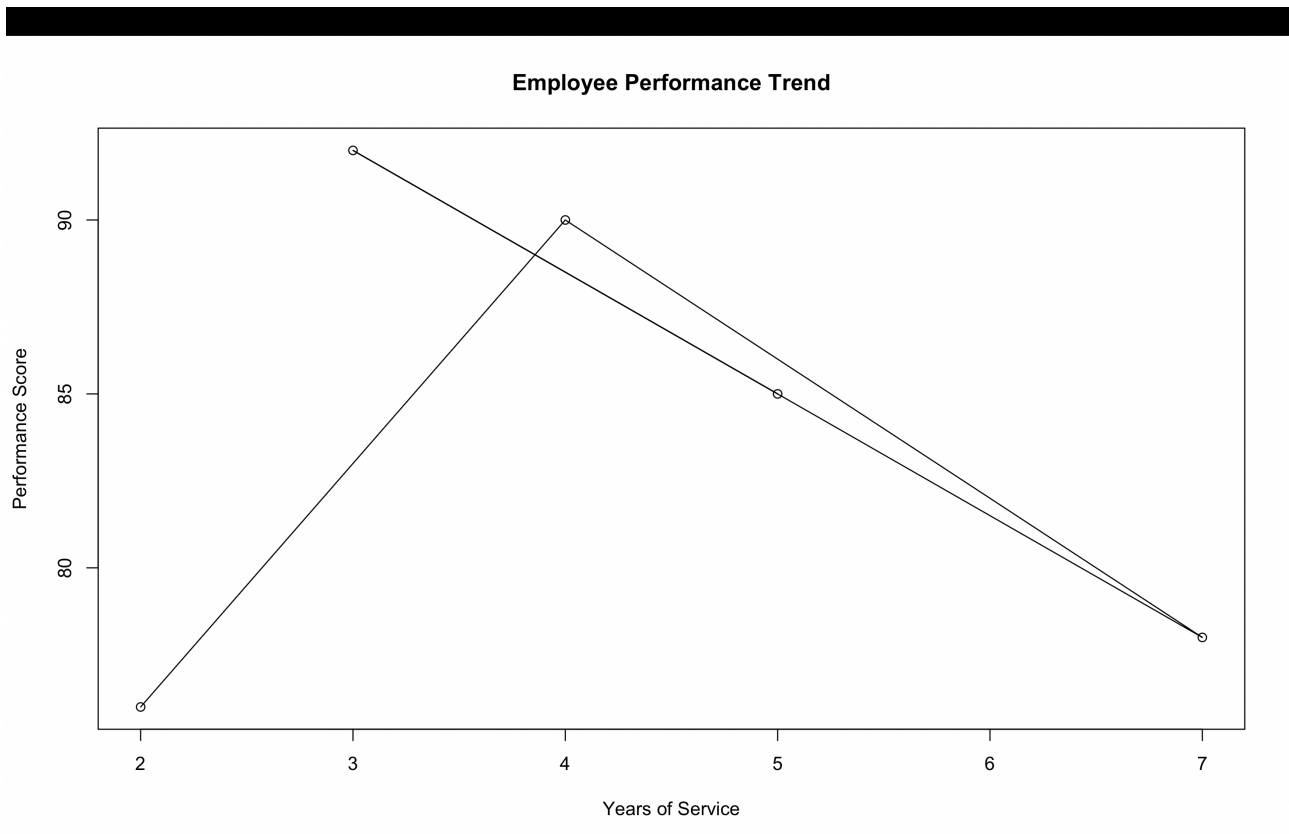


Q2. Employee Performance Trend Over Time

Program:

```
employee <- data.frame(
  id = c(1,2,3,4,5),
  department = c("Sales","HR","Marketing","Sales","HR"),
  years = c(5,3,7,4,2),
  performance = c(85,92,78,90,76)
)
plot(
  employee$years,
  employee$performance,
  type = "o",
  main = "Employee Performance Trend",
  xlab = "Years of Service",
  ylab = "Performance Score"
)
```

Output:



Q3. Employee Department Distribution and Performance Dashboard

Program:

```
employee <- data.frame(  
  id = c(1,2,3,4,5),  
  department = c("Sales","HR","Marketing","Sales","HR"),  
  years = c(5,3,7,4,2),  
  performance = c(85,92,78,90,76)  
)
```

```
par(mfrow = c(1, 2))
```

```
barplot(  
  table(employee$department),  
  main = "Department Distribution",  
  xlab = "Department",  
  ylab = "Employees"  
)
```

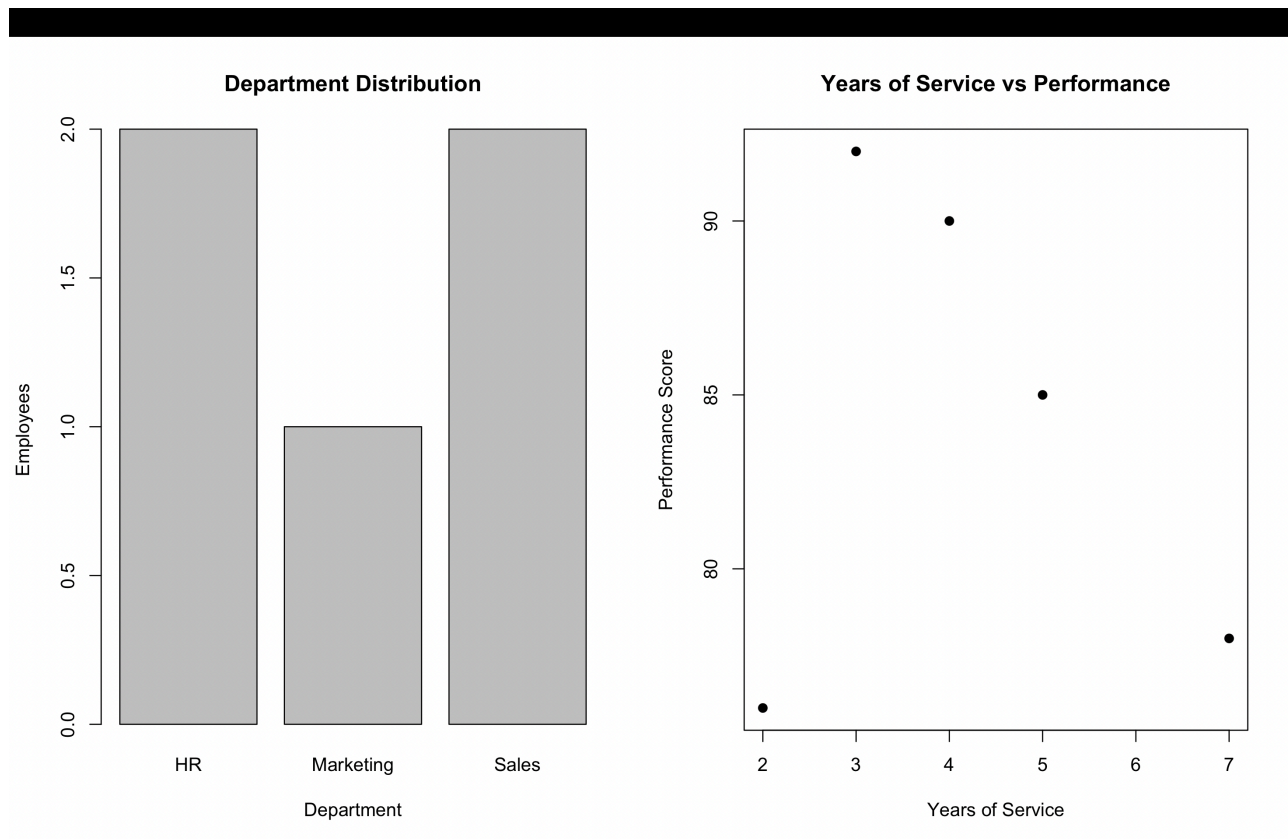
```
plot(  
  employee$years,  
  employee$performance,  
  main = "Years of Service vs Performance",  
  xlab = "Years of Service",
```

```

ylab = "Performance Score",
pch = 19
)

```

par(mfrow = c(1, 1))Output:



Q4. Employee Performance Data Table

Program:

```

employee <- data.frame(
  id = c(1,2,3,4,5),
  department = c("Sales","HR","Marketing","Sales","HR"),
  years = c(5,3,7,4,2),
  performance = c(85,92,78,90,76)
)
print(employee)

```

Output:

	id	department	years	performance
1	1	Sales	5	85
2	2	HR	3	92
3	3	Marketing	7	78
4	4	Sales	4	90
5	5	HR	2	76

> |

Set 24 Product Inventory Management

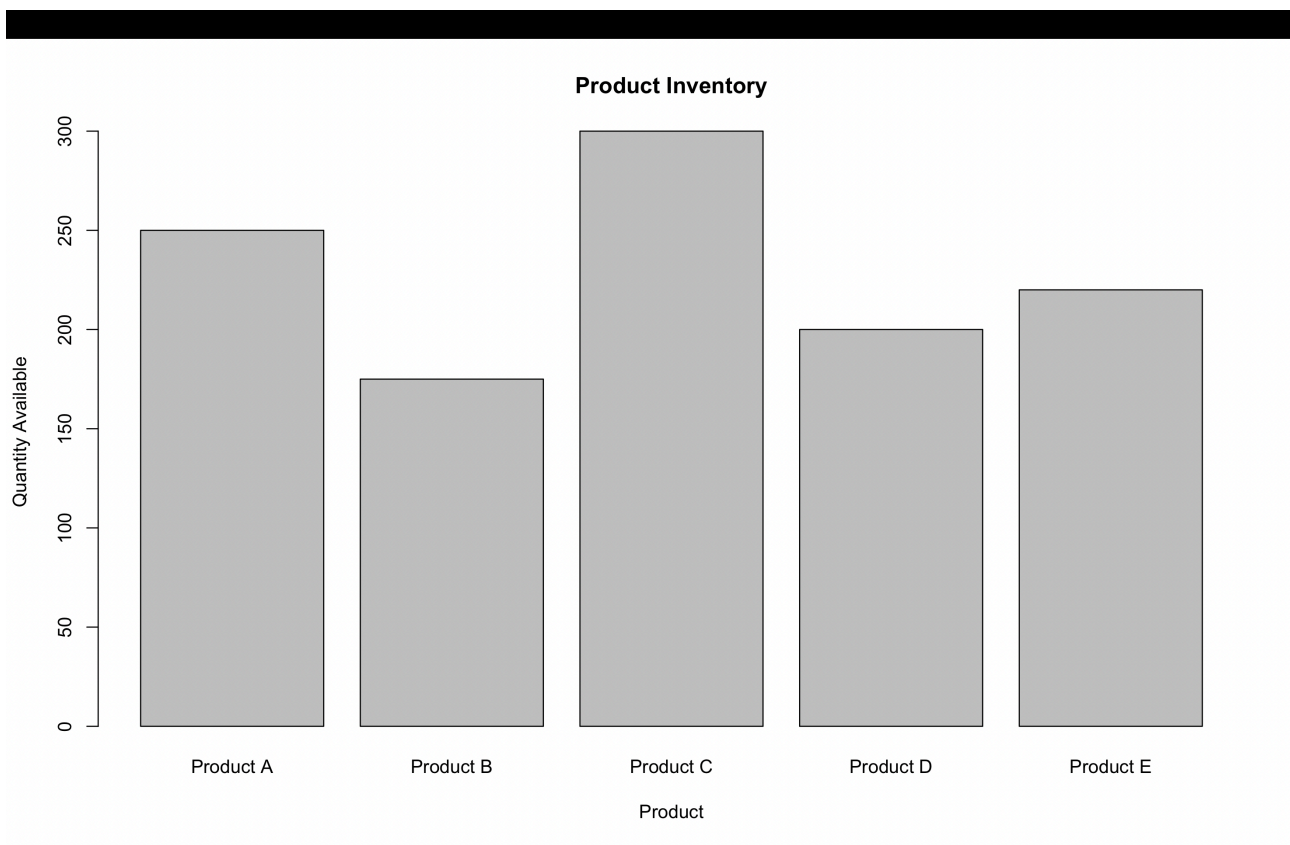
Q1. Quantity Available for Each Product

Program:

```
product <- c("Product A", "Product B", "Product C", "Product D", "Product E")
quantity <- c(250, 175, 300, 200, 220)

barplot(
  quantity,
  names.arg = product,
  main = "Product Inventory",
  xlab = "Product",
  ylab = "Quantity Available"
)
```

Output:



Q2. Stacked Bar Chart for Product Categories

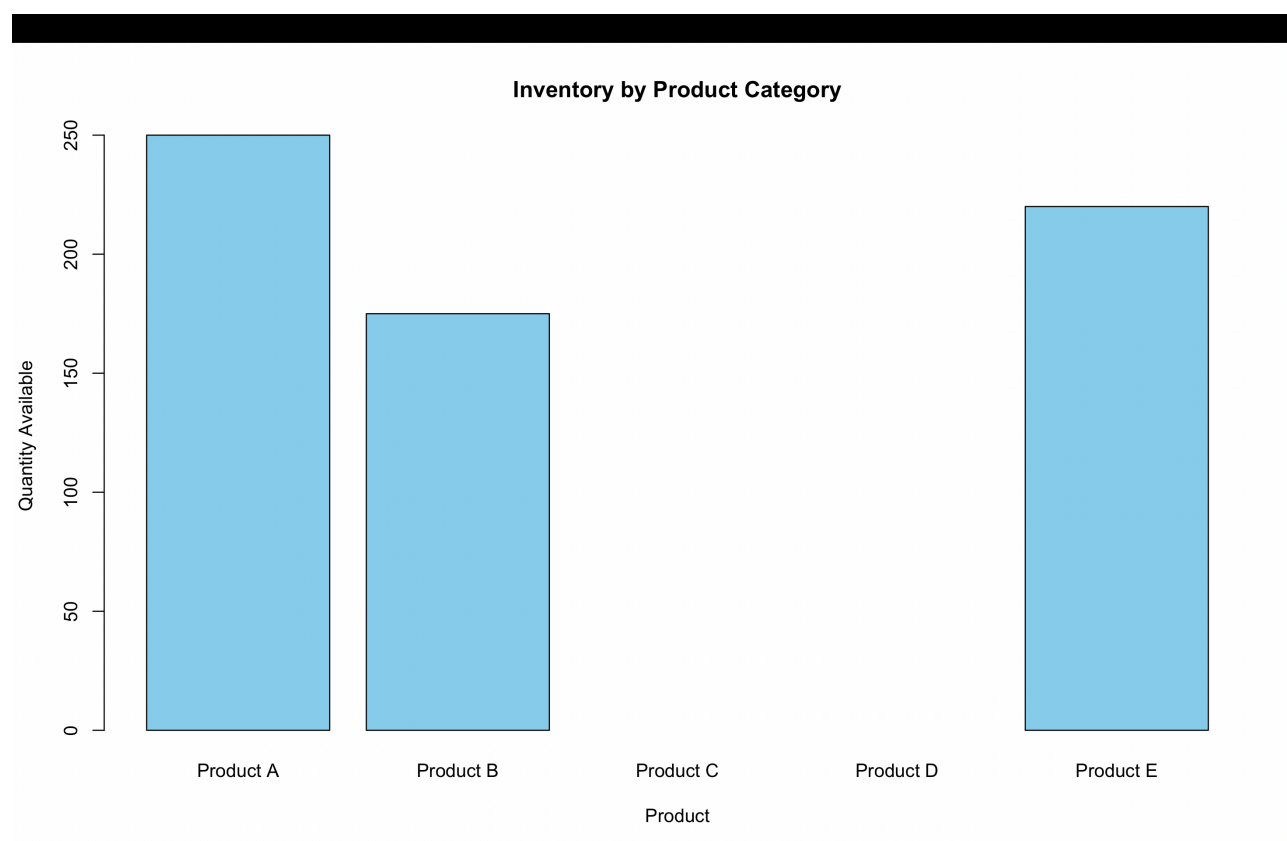
Program:

```
category <- c("Electronics", "Electronics", "Furniture", "Furniture", "Electronics")
```

```
data <- data.frame(
  product = product,
  category = category,
  quantity = quantity
)

barplot(
  tapply(data$quantity, list(data$category, data$product), sum),
  main = "Inventory by Product Category",
  xlab = "Product",
  ylab = "Quantity Available",
  col = c("skyblue", "lightgreen")
)
```

Output:



Q3. Product Inventory Dashboard

Program:

```
category <- c("Electronics", "Electronics", "Furniture", "Furniture", "Electronics")

data <- data.frame(
  product = product,
  category = category,
  quantity = quantity
)
```

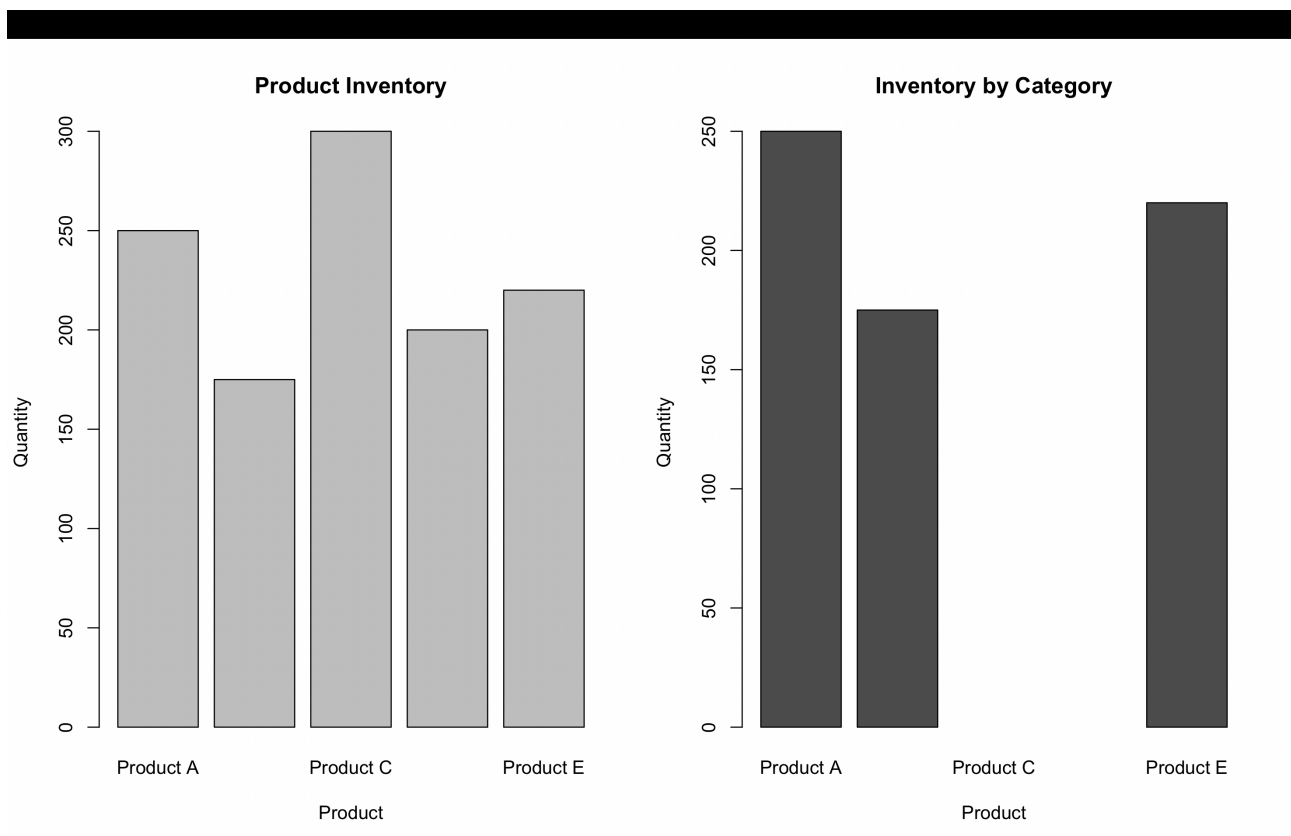
```
par(mfrow = c(1, 2))
```

```
barplot(  
  quantity,  
  names.arg = product,  
  main = "Product Inventory",  
  xlab = "Product",  
  ylab = "Quantity"  
)
```

```
barplot(  
  tapply(data$quantity, list(data$category, data$product), sum),  
  main = "Inventory by Category",  
  xlab = "Product",  
  ylab = "Quantity"  
)
```

```
par(mfrow = c(1, 1))
```

Output:



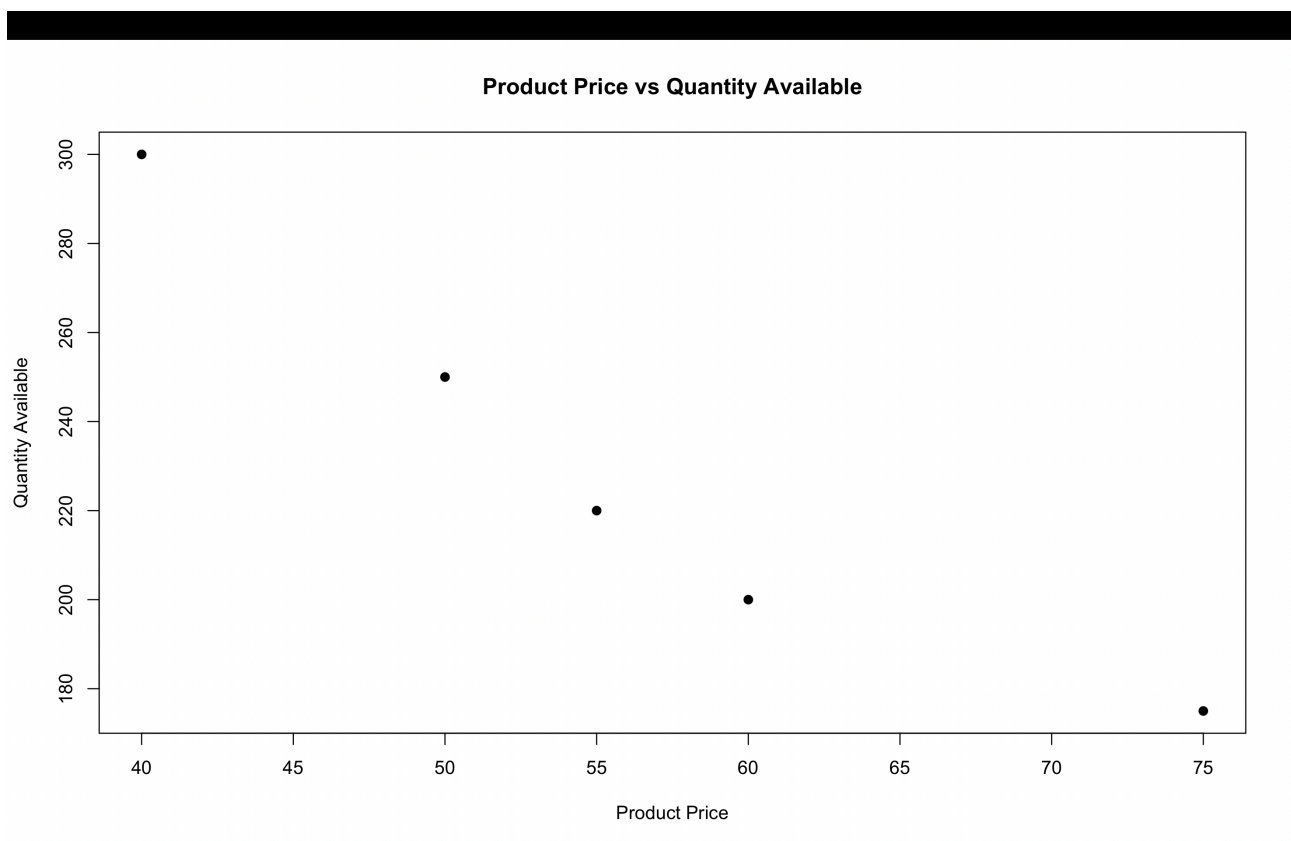
Q4. Product Price vs Quantity Available

Program:

```
price <- c(50, 75, 40, 60, 55)
```

```
plot(  
  price,  
  quantity,  
  main = "Product Price vs Quantity Available",  
  xlab = "Product Price",  
  ylab = "Quantity Available",  
  pch = 19  
)
```

Output:



Set 25: Website Traffic Analysis

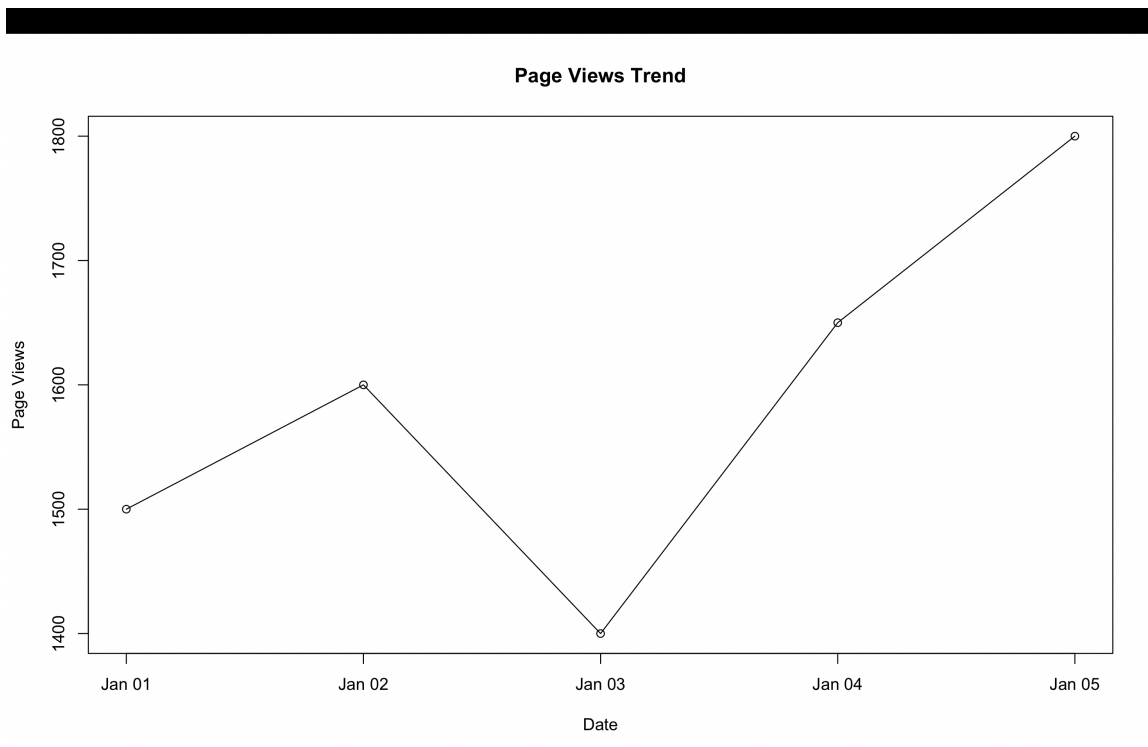
Q1. Website Page Views Trend

Program:

```
date <- as.Date(c("2023-01-01", "2023-01-02", "2023-01-03", "2023-01-04", "2023-01-05"))  
page_views <- c(1500, 1600, 1400, 1650, 1800)
```

```
plot(
  date,
  page_views,
  type = "o",
  main = "Page Views Trend",
  xlab = "Date",
  ylab = "Page Views"
)
```

Output:



Q2. Top N Days with Highest Click-through Rates

Program:

```
date <- as.Date(c("2023-01-01", "2023-01-02", "2023-01-03", "2023-01-04", "2023-01-05"))
ctr <- c(2.3, 2.7, 2.0, 2.4, 2.6)
```

```
n <- 3
```

```
top <- order(ctr, decreasing = TRUE)[1:n]
```

```
barplot(
  ctr[top],
  names.arg = date[top],
  main = "Top Days by Click-through Rate",

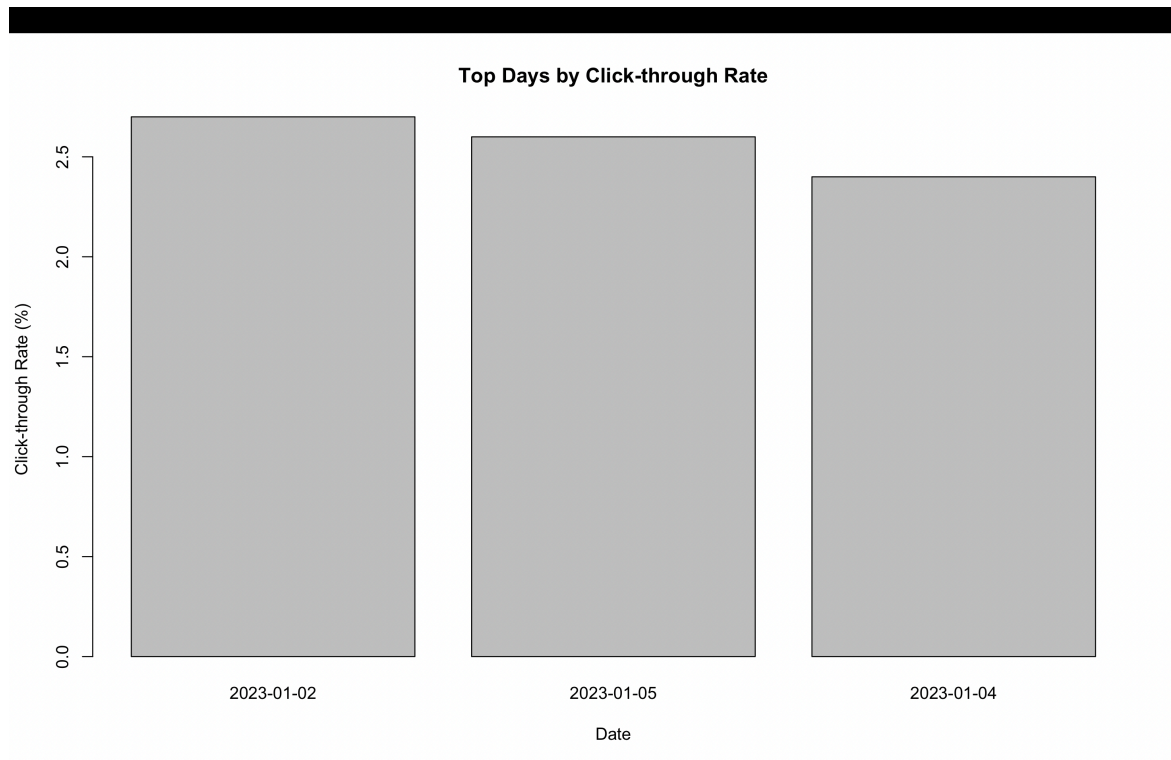
```

```

xlab = "Date",
ylab = "Click-through Rate (%)"
)

```

Output:



Q3. Stacked Area Chart of User Interactions

Program:

```

date <- as.Date(c("2023-01-01", "2023-01-02", "2023-01-03", "2023-01-04", "2023-01-05"))

```

```

likes <- c(300, 350, 280, 400, 450)

```

```

shares <- c(100, 120, 90, 150, 180)

```

```

comments <- c(50, 60, 45, 70, 80)

```

```

plot(
  date,
  likes,
  type = "n",
  main = "User Interactions",
  xlab = "Date",
  ylab = "Interactions"
)

```

```

polygon(
  c(date, rev(date)),

```

```

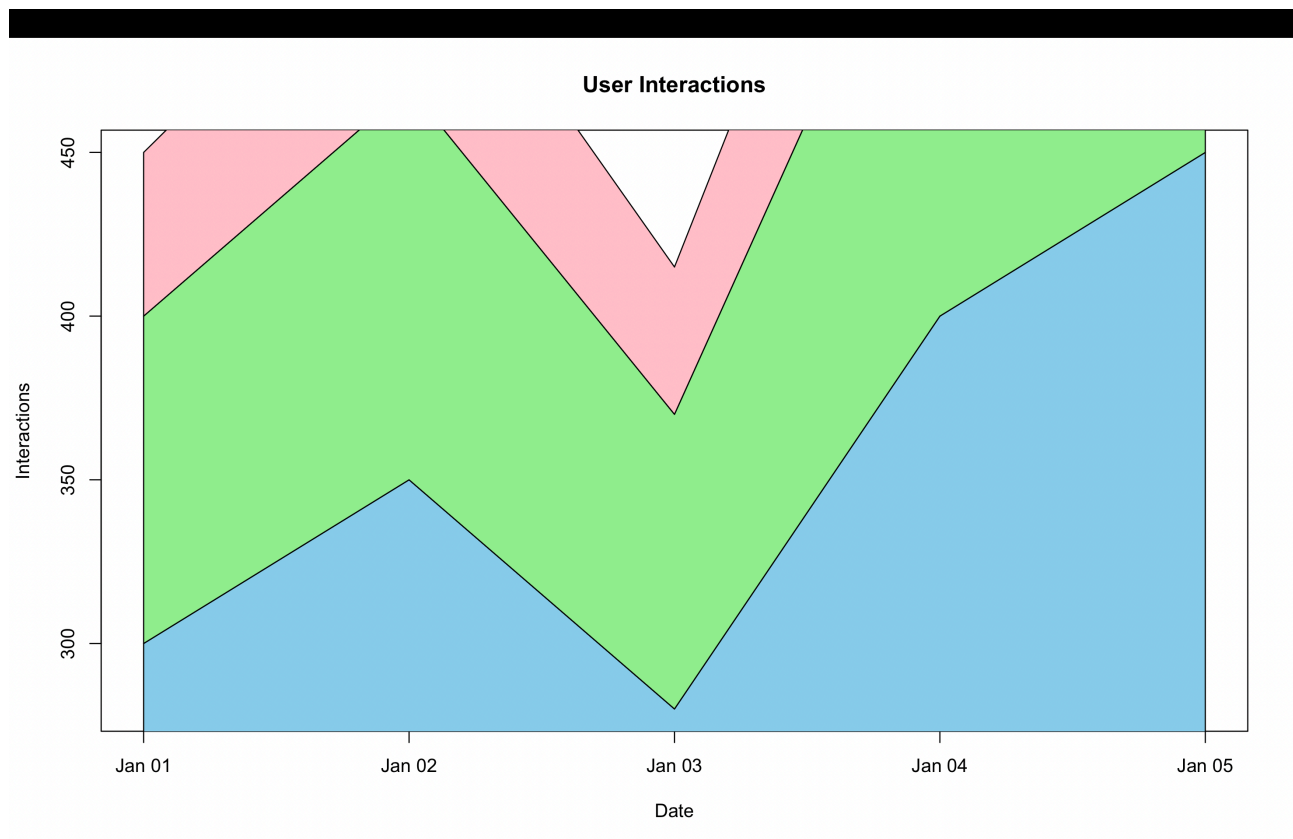
c(likes, rep(0, length(likes))),
col = "skyblue"
)

polygon(
  c(date, rev(date)),
  c(likes + shares, rev(likes)),
  col = "lightgreen"
)

polygon(
  c(date, rev(date)),
  c(likes + shares + comments, rev(likes + shares)),
  col = "pink"
)

```

Output:



Q4. Traffic Sources Map and Page Views Dashboard

Program:

```

source <- c("Google", "Facebook", "Instagram", "YouTube")
page_views <- c(3200, 2500, 1800, 1500)

barplot(
  page_views,

```

```
names.arg = source,  
main = "Page Views by Traffic Source",  
xlab = "Traffic Source",  
ylab = "Page Views"  
)
```

Output:

